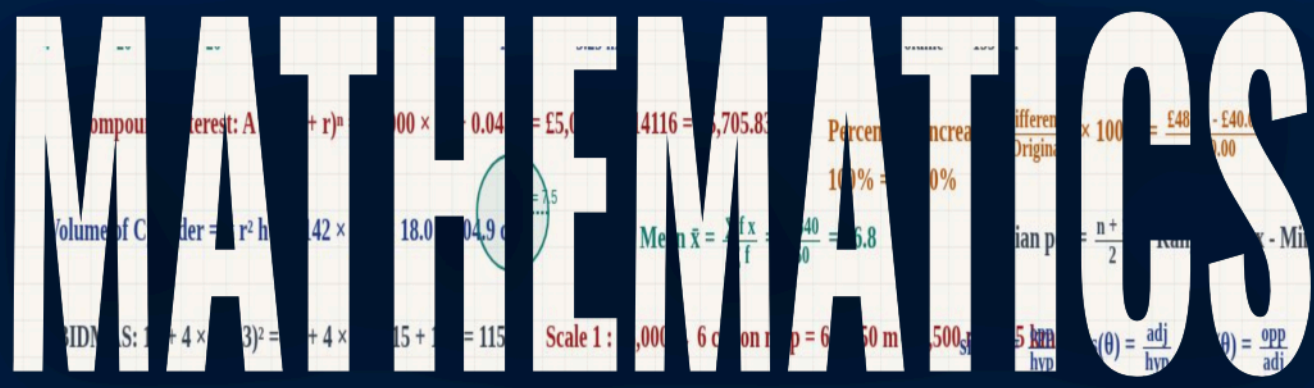
MASTER FUNCTIONAL SKILLS

FUNCTIONAL SKILLS MATHEMATICS

CORE REVISION GUIDE • MODEL ANSWERS

LEVEL 2



MATHEMATICS
CORE STUDY GUIDE

Complete 39-standard curriculum guide with Section A & Section B model solutions

39 Standards | Sec A Non-Calc | Sec B Calculator | Worked Solutions

OFQUAL REGULATED SPECIFICATION • LEVEL 2

OFFICIAL STUDY MANUAL

MASTER FUNCTIONAL SKILLS · REFORMED MATHEMATICS CURRICULUM DIRECTORY

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Master Functional Skills • **Mathematics Manual** • Number, Measures, Shape, Space & Data Handling **Lead Tutor:** Abdul • **Edition:** 2026/2027 Reformed Standards

NUMBER • STANDARD N1

1 Place Value, Ordering & Large Numbers



Topic 1 Applied Context: Place Value, Ordering & Large Numb...

★ Core Formula & Statutory Method

In large numbers up to 1,000,000, each digit column represents a power of ten. Use commas or spaces between thousands.

Separate digits into groups of three from the right: Millions, Thousands, Units.

✎ Official Worked Solution

A warehouse records 425,800 units in Q1 and 389,950 units in Q2. How many more in Q1?

1. Align columns: 425,800 - 389,950
2. Perform column subtraction with borrowing
3. Difference = **35,850 units**

✓ Examiner Golden Rule

Always align digits by their place value, never by the leftmost digit.

! Watch Out!

Don't confuse hundred thousands with millions.

TOPIC 1 • CANDIDATE WORKSPACE

P

Place Value, Ordering & Large Numbers: Practice Drills

1 **Question 1 (Foundation):** Write in figures: 'Six hundred and forty-two thousand, and eighty-five'. Then order these populations from smallest to largest: 485,200; 458,900; 485,090; 502,100. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A regional distribution hub processed 425,800 parcels in Q1 and 389,950 parcels in Q2. How many more parcels were processed in Q1 than in Q2? **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A council has an annual housing budget of £1,250,000. It spends £485,600 on roof repairs, £392,450 on heating upgrades, and £184,200 on external insulation. How much budget remains unspent? **(4 marks)**

Show your working

Answer: _____

NUMBER • STANDARD N2

2 Negative Numbers & Financial Balances



Topic 2 Applied Context: Negative Numbers & Financial Balan...

★ Core Formula & Statutory Method

Negative numbers represent values below zero (bank overdrafts, temperatures, sea level). Adding a negative is subtraction.

Positive + Positive = Positive. Positive + Negative = Subtract smaller from larger.

✎ Official Worked Solution

A business account is at -£1,450. A client pays £3,200, then bills of £850 and £1,120 are debited.

1. Start: -£1,450 + £3,200 = +£1,750
2. Deduct bills: £1,750 - £850 = £900
3. Deduct staff: £900 - £1,120 = **-£220** (Overdraft)

✓ Examiner Golden Rule

Moving right on the number line increases value; moving left decreases value.

! Watch Out!

Subtracting a negative becomes an addition: 5 - (-3) = 8.

TOPIC 2 • CANDIDATE WORKSPACE

P Negative Numbers & Financial Balances: Practice Drills

1 **Question 1 (Foundation):** Order these values from coldest to warmest: -12°C, +3°C, -4.5°C, -0.5°C, +1.8°C, -15°C. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A warehouse chiller is set at +3°C. A fault causes the temperature to drop by 8.5°C. What is the new temperature? If the safety alarm triggers at -5°C, does the alarm sound? **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A small business starts Monday with a bank balance of -£640. Over the week: sales revenue of +£2,850 is paid in, direct debits of £1,420 and £980 clear, and a supplier refund of £195 arrives. What is the Friday closing balance? **(4 marks)**

Show your working

Answer: _____

NUMBER • STANDARD N3

3 Fraction Operations & Mixed Numbers



Topic 3 Applied Context: Fraction Operations & Mixed Numbers

★ Core Formula & Statutory Method

Fractions represent equal parts of a whole. To add or subtract, you MUST find a common denominator.

Multiply fractions: multiply numerators together, then denominators together. Divide: flip second fraction and multiply.

✎ Official Worked Solution

Calculate 2 1/4 + 1 2/3 for a catering recipe calculation.

1. Convert to improper: 9/4 + 5/3
2. Common denominator 12: 27/12 + 20/12 = 47/12
3. Convert to mixed: **3 11/12**

✓ Examiner Golden Rule

Always convert mixed numbers to improper fractions before multiplying or dividing.

! Watch Out!

Never add denominators together: 1/2 + 1/2 is not 2/4!

TOPIC 3 • CANDIDATE WORKSPACE

P

Fraction Operations & Mixed Numbers: Practice Drills

1 **Question 1 (Foundation):** Evaluate giving your answers in simplest form: a) 5/8 + 1/6 b) 2 2/3 × 3/4. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A chef has 4 1/2 kg of prime beef mince. Each burger patty requires 3/16 kg of mince. How many complete burger patties can the chef make? **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** In a manufacturing plant, 3/5 of workers are on day shift, 1/4 are on late shift, and the remaining 45 workers are on night shift. How many total workers are employed at the plant? **(4 marks)**

Show your working

Answer: _____

NUMBER • STANDARD N4

4 Percentages of Amounts & Multipliers



Topic 4 Applied Context: Percentages of Amounts & Multipliers

★ Core Formula & Statutory Method

Percent means 'out of 100'. Use single-step decimal multipliers to calculate percentage changes efficiently.

Percentage increase: Multiplier = 1 + (percentage / 100). Decrease: Multiplier = 1 - (percentage / 100).

✎ Official Worked Solution

An operative earns £24,500. They receive a 4.5% cost of living pay rise. What is the new salary?

1. Multiplier for 4.5% increase = 1.045
2. Calculation: £24,500 x 1.045
3. New salary = **£25,602.50**

✓ Examiner Golden Rule

Use decimal multipliers for calculator questions to save time and reduce errors.

! Watch Out!

Check if question asks for the INCREASE or the FINAL TOTAL.

TOPIC 4 • CANDIDATE WORKSPACE

P

Percentages of Amounts & Multipliers: Practice Drills

1 **Question 1 (Foundation):** Write the single decimal multiplier for: **(2 marks)**
a) Finding 6.5% b) Increasing by 12% c) Decreasing by 35% d) Decreasing by 4%.

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A car insurance premium of £620.00 increases by 8.5% at renewal. What is the new renewal price? **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A distribution depot ships 14,500 parcels in November. In December, parcel volume increases by 28%. In January, volume drops by 15% compared to December. How many parcels are shipped in January? **(4 marks)**

Show your working

Answer: _____

NUMBER • STANDARD N5

5 Reverse Percentages & Original Value



Topic 5 Applied Context: Reverse Percentages & Original Value

★ Core Formula & Statutory Method

Reverse percentage problems give you the value AFTER an increase or decrease and ask for the ORIGINAL value.

Original Value = Final Value / Multiplier. NEVER just calculate the percentage of the final value!

✎ Official Worked Solution

A laptop is on sale for £468 after a 20% discount. What was the original price before the sale?

1. After 20% off, £468 represents 80% of original price (0.80 multiplier)
2. Calculation: £468 / 0.80
3. Original price = **£585.00**

✓ Examiner Golden Rule

Always divide by the percentage multiplier to work backwards to the original 100%.

! Watch Out!

Do NOT add 20% of £468 (£93.60) — that gives the wrong answer (£561.60).

TOPIC 5 • CANDIDATE WORKSPACE

P

Reverse Percentages & Original Value: Practice Drills

1 **Question 1 (Foundation):** A power drill sells for £84.00 including 20% VAT. Calculate the price before VAT was added. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A bakery sold 450 loaves on Friday and 558 loaves on Saturday. Calculate the percentage increase in sales. **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A fitness coach bought commercial gym equipment for £3,200. Three years later, he sells it for £1,840. Calculate the percentage loss in value. If a new model is currently advertised at £2,760 after a 15% discount, what was its original price? **(4 marks)**

Show your working

Answer: _____

NUMBER • STANDARD N6

6 Compound Interest & Depreciation



Topic 6 Applied Context: Compound Interest & Depreciation

★ Core Formula & Statutory Method

Compound interest applies interest to both the initial principal and the accumulated interest from previous periods.

Final Amount = Principal x (1 + r)^n, where r is decimal interest rate and n is number of years.

✎ Official Worked Solution

A healthcare worker invests £5,000 in a savings bond paying 3.2% compound interest per year for 3 years.

1. Multiplier = 1.032
2. Formula: £5,000 x (1.032)^3
3. £5,000 x 1.09911 = **£5,495.55**

✓ Examiner Golden Rule

Apply the multiplier repeatedly or use the power key (x^y) on your scientific calculator.

! Watch Out!

Simple interest gives less than compound interest over multiple years.

TOPIC 6 • CANDIDATE WORKSPACE

P

Compound Interest & Depreciation: Practice Drills

1 **Question 1 (Foundation):** Calculate the value of £2,400 invested for 2 years at 3% compound interest per annum. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A commercial delivery van was purchased brand new for £28,000. It depreciates at a rate of 18% per year. What is its estimated book value after 3 years? **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** An investor has £10,000 to invest for 3 years. Bank A offers 4.5% simple interest per year. Bank B offers 4.2% compound interest per annum. Which bank will yield more total interest at the end of 3 years, and by how much? **(4 marks)**

Show your working

Answer: _____

7

NUMBER • STANDARD N7

Ratio Division in 2 & 3 Parts



Topic 7 Applied Context: Ratio Division in 2 & 3 Parts

Core Formula & Statutory Method

Ratios show relative sizes of quantities. To divide into a ratio, find the value of one part.

Step 1: Add ratio parts. Step 2: Total amount / Total parts = 1 part. Step 3: Multiply by each ratio part.

Official Worked Solution

Concrete is mixed in the ratio cement : sand : gravel = 1 : 2 : 4. How much sand is needed for 840 kg of mix?

1. Total parts: 1 + 2 + 4 = 7 parts
2. Value of 1 part: 840 kg / 7 = 120 kg
3. Sand (2 parts): 2 x 120 kg = **240 kg**

Examiner Golden Rule

Always check your answers sum back to the original total (120 + 240 + 480 = 840).

Watch Out!

Read the order of terms carefully: cement : sand : gravel.

TOPIC 7 • CANDIDATE WORKSPACE

P

Ratio Division in 2 & 3 Parts: Practice Drills

1 **Question 1 (Foundation):** Share £4,500 in the ratio 5 : 3 : 1. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A mortar mix uses cement, sand, and gravel in the ratio 1 : 2 : 4. A builder needs 1.4 tonnes of mortar. How many kilograms of sand does he require? **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** Supermarket A sells a 750 g box of cereal for £2.85. Supermarket B sells a 1.2 kg family pack for £4.32. Which pack offers better value for money? Justify your decision with full unit calculations. **(4 marks)**

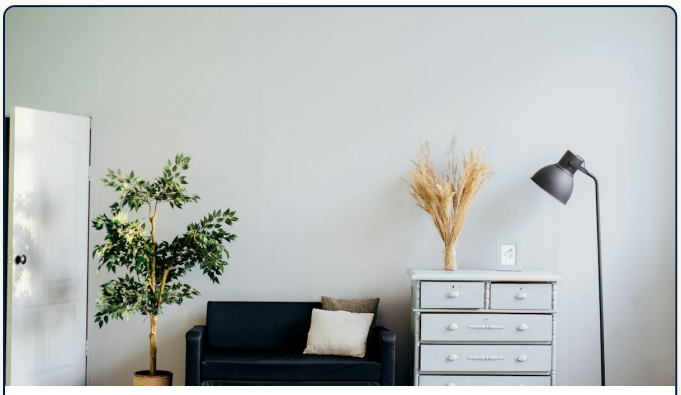
Show your working

Answer: _____

8

NUMBER • STANDARD N8

Direct & Inverse Proportion



Topic 8 Applied Context: Direct & Inverse Proportion

Core Formula & Statutory Method

Direct proportion: as one increases, the other increases at the same rate. Inverse: as one increases, the other decreases.

Direct: find unitary value (value for 1) then multiply. Inverse: Total work = workers x days = constant.

Official Worked Solution

It takes 4 decorators 6 days to paint an office block. How long would it take 3 decorators?

1. Total work required = 4 decorators x 6 days = 24 worker-days
2. Time for 3 decorators = 24 worker-days / 3 decorators
3. Total time = **8 days**

Examiner Golden Rule

More workers means FEWER days: inverse proportion always multiplies to a constant product.

Watch Out!

Direct proportion problems maintain a constant ratio (division).

TOPIC 8 • CANDIDATE WORKSPACE

P Direct & Inverse Proportion: Practice Drills

1 **Question 1 (Foundation):** It takes 5 cleaners 4 hours to clean an office floor. How long would it take 8 cleaners working at the exact same rate? **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A pipe delivers 45 litres/minute and fills a water storage tank in 80 minutes. How long would it take to fill the tank if the flow rate was reduced to 30 litres/minute? **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A groundworks contractor estimates that 3 excavators can clear a development site in 16 working days. The client requests the work be completed in 6 working days. How many additional excavators must the contractor hire? **(4 marks)**

Show your working

Answer: _____

NUMBER • STANDARD N9

9 Decimals & Money Calculations



Topic 9 Applied Context: Decimals & Money Calculations

★ Core Formula & Statutory Method

Decimals represent tenths, hundredths, and thousandths. Money is always rounded to 2 decimal places (£0.00).

Align decimal points vertically for addition and subtraction. For multiplication, count total decimal places.

✎ Official Worked Solution

An operative buys 14.5 metres of cabling at £2.64 per metre plus VAT at 20%.

1. Cable cost: 14.5 x £2.64 = £38.28
2. Add 20% VAT: £38.28 x 1.20 = £45.936
3. Rounded to 2 decimal places = **£45.94**

✓ Examiner Golden Rule

Always include two decimal places for currency answers (write £45.90, not £45.9).

! Watch Out!

Round down only at the very final step of your multi-step calculation.

TOPIC 9 • CANDIDATE WORKSPACE

P Decimals & Money Calculations: Practice Drills

1 **Question 1 (Foundation):** Evaluate without a calculator: a) 4.85 × 100 b) 73.6 ÷ 1000 c) 14.6 - 8.79. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A dental receptionist checks three utility payments: electricity £184.25, gas £112.80, and water £46.45. If the practice had budgeted £350.00, by how much did they overspend or underspend? **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** An electrician quotes a client: 6 light fittings at £24.50 each, 45 m cable at £1.20/m, and 3.5 hours labour at £35/hr. The electrician offers a 10% prompt-payment discount on the total before VAT, and then adds 20% VAT. Calculate the final bill. **(4 marks)**

Show your working

Answer: _____

NUMBER • STANDARD N10

10 Order of Operations (BIDMAS)



Topic 10 Applied Context: Order of Operations (BIDMAS)

Core Formula & Statutory Method

Calculations must follow the statutory mathematical order: Brackets, Indices, Division/Multiplication, Addition/Subtraction.

Work inside brackets first, then evaluate powers/roots, then left-to-right for division/multiplication.

Official Worked Solution

Evaluate the formula for monthly mortgage repayment factor: (180 - 4 x 5^2) / 2.

1. Indices first: 5^2 = 25
2. Multiplication: 4 x 25 = 100
3. Brackets: 180 - 100 = 80
4. Division: 80 / 2 = **40**

Examiner Golden Rule

Calculators without bracket keys do operations from left to right; enter BIDMAS steps manually.

Watch Out!

Division and multiplication have equal priority — execute left to right.

TOPIC 10 • CANDIDATE WORKSPACE

P Order of Operations (BIDMAS): Practice Drills

1 **Question 1 (Foundation):** Evaluate without a calculator: a) 15 - 3 × 4 + 2 b) (8 + 4) ÷ (7 - 3) + 3^2. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** Engineer's braking distance formula: $d = 0.05v^2 + 0.4v$, where v is speed in mph. Calculate the stopping distance d when v = 40 mph. **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** Total hire charge C (in £) for an industrial generator is given by C = 1.20 × (65 + 35d - 20), where d is the number of days hired. Calculate C for an 8-day hire. **(4 marks)**

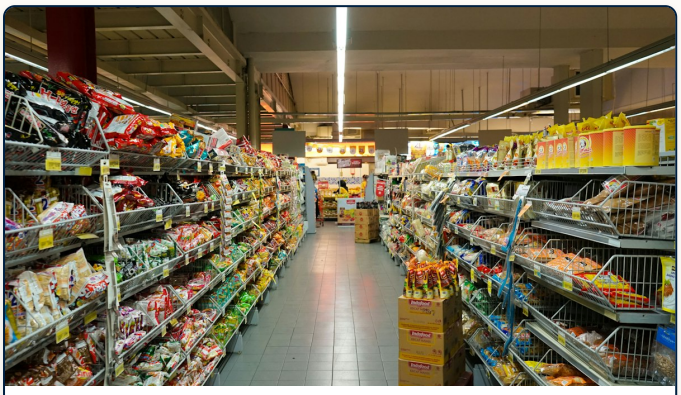
Show your working

Answer: _____

11

NUMBER • STANDARD N11

Estimation & Approximation



Topic 11 Applied Context: Estimation & Approximation

Core Formula & Statutory Method

Estimation involves rounding every number in a calculation to 1 significant figure to check plausibility.

Round each number to 1 significant figure BEFORE performing any calculation.

Official Worked Solution

Estimate the cost of 48.7 hours of plant machinery hire at £31.25 per hour.

1. Round 48.7 to 1 sig fig = 50
2. Round £31.25 to 1 sig fig = £30
3. Estimated calculation = 50 x £30 = **£1,500**

Examiner Golden Rule

Do not calculate the exact answer and then round it — examiners award marks for pre-rounding.

Watch Out!

Estimation is an essential checking method required in Section A exam tasks.

P

TOPIC 11 • CANDIDATE WORKSPACE

Estimation & Approximation: Practice Drills

1 **Question 1 (Foundation):** Round each number to 1 significant figure to estimate: a) 48.7 × 31.2 b) 812 ÷ 19.6. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A warehouse manager orders 195 heavy-duty crates at £48.75 each. Use 1 significant figure estimation to find an approximate total cost, and state whether it is an overestimate or underestimate. **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A parcel courier measures a crate as 60 cm by 40 cm by 30 cm, each rounded to the nearest 10 cm. Calculate the upper bound (maximum possible volume) of the crate in cm^3. **(4 marks)**

Show your working

Answer: _____

MEASURE • STANDARD M12

12 Metric & Imperial Conversions



Topic 12 Applied Context: Metric & Imperial Conversions

★ Core Formula & Statutory Method

Functional skills requires converting between metric units (mm, cm, m, km; g, kg; ml, l) and key imperial benchmarks.

Key imperial equivalences: 1 inch = 2.54 cm; 1 foot = 30 cm; 1 kg = 2.2 lbs; 1 gallon = 4.5 litres; 5 miles = 8 km.

✎ Official Worked Solution

A patient weighs 14 stone 5 lbs in clinic. Convert this weight into kilograms.

1. Total pounds: (14 x 14) + 5 = 196 + 5 = 201 lbs
2. Convert lbs to kg: 201 / 2.2
3. Weight = **91.36 kg**

✓ Examiner Golden Rule

Check whether your conversion factor requires multiplying or dividing.

! Watch Out!

Remember: 1 stone = 14 pounds (lbs).

TOPIC 12 • CANDIDATE WORKSPACE

P Metric & Imperial Conversions: Practice Drills

1 **Question 1 (Foundation):** Convert: a) 3.45 kg into grams b) 750 ml into litres c) 140 miles into kilometres using 5 miles = 8 km. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A patient in hospital is prescribed liquid medication: 15 ml three times per day for 14 days. Will a 500 ml bottle be sufficient? Show full working. **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A UK haulage truck carries 14 tonnes of gravel. The European delivery limit is 31,000 lbs. Use 1 kg = 2.2 lbs to determine if the truck is within legal weight limits. **(4 marks)**

Show your working

Answer: _____

MEASURE • STANDARD M13

13 Compound Units: Speed, Density & Pressure



Topic 13 Applied Context: Compound Units: Speed, Density & ...

★ Core Formula & Statutory Method

Speed is how fast something travels: Speed = Distance ÷ Time. Density = Mass ÷ Volume. Pressure = Force ÷ Area.

Speed formula triangle: Put Distance (D) at the top, Speed (S) and Time (T) at the bottom. Cover S to get D ÷ T. Cover D to get S × T. Cover T to get D ÷ S.

✎ Official Worked Solution

A delivery van travels 145 miles from Birmingham to London in 2 hours 45 minutes. Find average speed.

1. Convert 45 mins to decimal hours: 45 ÷ 60 = 0.75 hours
2. Total time = 2.75 hours
3. Speed = 145 ÷ 2.75 = **52.7 mph**

✓ Examiner Golden Rule

Always convert minutes to decimal hours by dividing by 60 (2 hours 45 mins is 2.75h, NOT 2.45h!).

! Watch Out!

Density is measured in g/cm³ or kg/m³.

TOPIC 13 • CANDIDATE WORKSPACE

P Compound Units: Speed, Density & Pressure: Practice Drills

1 **Question 1 (Foundation):** A brass casting has a volume of 250 cm³ and a mass of 2,125 g. Calculate the density of the brass in g/cm³. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A hydraulic piston exerts a force of 1,800 N on an area of 0.04 m². Calculate the pressure exerted in N/m². **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A sales rep travels from Leeds to Birmingham (115 miles). He departs at 07:45 and arrives at 10:09. Due to roadworks, he had to stop for a 15-minute coffee break. Calculate his average driving speed while moving. **(4 marks)**

Show your working

Answer: _____

MEASURE • STANDARD M14

14 Perimeter of Composite 2D Shapes



Topic 14 Applied Context: Perimeter of Composite 2D Shapes

★ Core Formula & Statutory Method

Perimeter is the total boundary distance around the outside edge of a 2D shape.

Find all missing horizontal and vertical side lengths first, then sum all outer edges.

✎ Official Worked Solution

An L-shaped industrial yard has outer dimensions 18 m by 14 m with an inset section 7 m by 6 m.

1. Missing horizontal = 18 - 7 = 11 m
2. Missing vertical = 14 - 6 = 8 m
3. Perimeter = 18 + 14 + 7 + 6 + 11 + 8 = **64 m**

✓ Examiner Golden Rule

Do not include interior dashed boundary lines in perimeter totals.

! Watch Out!

Tick off each side as you sum it to avoid missing any edge.

TOPIC 14 • CANDIDATE WORKSPACE

P Perimeter of Composite 2D Shapes: Practice Drills

1 **Question 1 (Foundation):** An L-shaped office has overall length 12 m and overall width 9 m. Find the total perimeter of the room. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A rectangular warehouse yard measures 28 m by 18 m, with a 4 m wide vehicle entrance gate where no fencing is required. Security fencing costs £32.50 per metre. Calculate the cost of the fencing. **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A T-shaped exhibition stand has a top bar of 10 m by 3 m and a central stem of 4 m wide by 6 m long. Aluminium edging strip is sold in 2.5 m lengths at £14.00 per length. How many lengths are needed for the full outer perimeter, and what is the total cost? **(4 marks)**

Show your working

Answer: _____

15

MEASURE • STANDARD M15

Circumference of Circles



Topic 15 Applied Context: Circumference of Circles

Core Formula & Statutory Method

Circumference is the distance around the outside edge of a circle: Circumference = 3.142 × diameter.

If you are given the radius, double it first to get the diameter: diameter = 2 × radius.

Official Worked Solution

A circular roundabout has a diameter of 24 metres. Calculate its outer circumference.

1. Formula: Circumference = 3.142 × diameter
2. Calculation: 3.142 × 24 m
3. Circumference = **75.41 metres**

Examiner Golden Rule

For a semi-circle perimeter: calculate (3.142 × diameter ÷ 2) + diameter (don't forget to add the flat straight edge!).

Watch Out!

Remember: Circumference = 3.142 × diameter.

TOPIC 15 • CANDIDATE WORKSPACE

P

Circumference of Circles: Practice Drills

1 **Question 1 (Foundation):** Calculate the circumference of a circular water tank with a radius of 4.5 metres. (Use π = 3.14) **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A running track consists of a rectangle measuring 100 m by 64 m with two semicircular ends. Calculate the total running distance around the outer edge of one complete lap. **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A decorative wrought-iron garden gate has a rectangular base (1.2 m wide by 1.8 m high) topped with a semicircular arch. What total length of iron framing is needed for the perimeter? (Use π = 3.142) **(4 marks)**

Show your working

Answer: _____

MEASURE • STANDARD M16

16 Area of Rectangles, Triangles & Parallelograms



Topic 16 Applied Context: Area of Rectangles, Triangles & Par...

★ Core Formula & Statutory Method

Area is the flat surface space inside a 2D shape, measured in square units (cm², m²).

Rectangle = length × width. Triangle = (base × height) ÷ 2. Parallelogram = base × height.

✎ Official Worked Solution

A triangular warehouse roof gable has a base of 8.4 m and perpendicular vertical height of 3.5 m.

1. Formula: Triangle Area = (base × height) ÷ 2
2. Calculation: (8.4 × 3.5) ÷ 2 = 29.4 ÷ 2
3. Area = **14.7 m²**

✓ Examiner Golden Rule

Always use the upright vertical height at 90°, NEVER the slanted sloping side!

! Watch Out!

For triangles, always divide by 2 at the end.

TOPIC 16 • CANDIDATE WORKSPACE

P Area of Rectangles, Triangles & Parallelograms: Practice Drills

1 **Question 1 (Foundation):** Calculate the area of: a) A triangle with base 14 cm and perpendicular height 9 cm b) A parallelogram with base 8.5 m and perpendicular height 4.0 m. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A rectangular gable wall is 8 m wide and 5 m high, topped by a triangular roof section of base 8 m and vertical height 2.5 m. Calculate the total exterior wall area to be rendered. **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A rectangular hallway measures 12 m by 3.5 m. Square ceramic floor tiles measure 50 cm by 50 cm and are sold in boxes of 8 tiles for £22.00 per box. Adding 10% extra tiles for cutting wastage, calculate the total cost of the boxes required. **(4 marks)**

Show your working

Answer: _____

MEASURE • STANDARD M17

17 Area of Trapeziums & Composite Shapes



Topic 17 Applied Context: Area of Trapeziums & Composite Sh...

★ Core Formula & Statutory Method

Trapezium area = ((a + b) ÷ 2) × h, where a and b are the parallel sides and h is the upright height.

Split complex composite shapes into separate rectangles and triangles, find each area, then add together.

✎ Official Worked Solution

A car park space is a trapezium with parallel sides 4.5 m and 6.5 m and perpendicular width 2.8 m.

1. Sum of parallel sides: 4.5 + 6.5 = 11.0 m
2. Average: 11.0 ÷ 2 = 5.5 m
3. Area: 5.5 × 2.8 = **15.4 m²**

✓ Examiner Golden Rule

Add the two parallel edges first, divide by 2, then multiply by the perpendicular distance.

! Watch Out!

For L-shapes, draw a dashed line and label sections 'Shape A' and 'Shape B'.

TOPIC 17 • CANDIDATE WORKSPACE

P Area of Trapeziums & Composite Shapes: Practice Drills

1 **Question 1 (Foundation):** Calculate the area of a circular patio with a diameter of 6 metres. Give your answer to 2 decimal places. (Use π = 3.142) **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A flat roof in the shape of a trapezium has parallel sides of 7.2 m and 11.8 m, and a perpendicular width of 5.0 m. Roofing felt costs £8.50 per square metre. Calculate the cost to felt the roof. **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A sports hall floor has an L-shaped layout: 14 m overall length by 10 m overall width. A cutout corner measures 6 m by 4 m. A tin of floor varnish covers 25 m² and costs £34.99. How many tins must be bought, and what is the total cost? **(4 marks)**

Show your working

Answer: _____

18

MEASURE • STANDARD M18

Area of Circles & Semicircles



Topic 18 Applied Context: Area of Circles & Semicircles

Core Formula & Statutory Method

Area of a circle: Circle Area = π r^2 = 3.142 × r^2 (where the ² symbol means squared: r^2 = radius × radius). Multiply the radius by itself (r^2) first, then multiply by 3.142.

If you are given the diameter, halve it first to get the radius: radius = diameter ÷ 2. Semicircle area = (π r^2) ÷ 2 = (3.142 × r^2) ÷ 2.

Official Worked Solution

A circular patio has a diameter of 7.2 metres. Calculate its total paved surface area.

1. Halve diameter to get radius: 7.2 ÷ 2 = 3.6 m
2. Square radius (r^2 = r × r): 3.6^2 = 3.6 × 3.6 = 12.96
3. Area: π r^2 = 3.142 × 12.96 = **40.72 m^2**

Examiner Golden Rule

Always square the radius (r^2 = radius × radius) before multiplying by 3.142.

Watch Out!

Do NOT use diameter in the area calculation — always halve to radius first!

TOPIC 18 • CANDIDATE WORKSPACE

P Area of Circles & Semicircles: Practice Drills

1 **Question 1 (Foundation):** Calculate the area of a circular patio with a diameter of 6 metres. Give your answer to 2 decimal places. (Use π = 3.142) **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A semicircular lawn has a straight diameter edge of 10 metres. Lawn turf costs £5.80 per m². Calculate the cost to turf the semicircular lawn. (Use π = 3.142) **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A rectangular park area measuring 20 m by 14 m contains a circular ornamental pond of radius 4 m. The area outside the pond is covered in gravel at 40 kg per m². How many 800 kg bulk bags of gravel must be ordered? (Use π = 3.142) **(4 marks)**

Show your working

Answer: _____

MEASURE • STANDARD M19

19 Surface Area of 3D Prisms & Cylinders



Topic 19 Applied Context: Surface Area of 3D Prisms & Cylind...

★ Core Formula & Statutory Method

Total surface area is the sum of the areas of all external faces of a 3D solid.

Cuboid: $2(lw + lh + wh)$. Cylinder: $2\pi r^2 + \pi dh = (2 \times 3.142 \times r^2) + (3.142 \times d \times h)$, where r^2 means r × r [two circular ends + curved rectangle].

✎ Official Worked Solution

A cylindrical storage water tank has radius 1.5 m and height 4 m. Calculate total surface area.

1. Two circular ends: $2 \times \pi\ r^2 = 2 \times 3.142 \times 1.5^2 = 2 \times 3.142 \times 2.25 = 14.14\ m^2$
2. Curved side: $\pi \times d \times h = 3.142 \times 3 \times 4 = 37.70\ m^2$
3. Total surface area = 14.14 + 37.70 = **51.84 m²**

✓ Examiner Golden Rule

The ² symbol means squared ($r^2 = r \times r$). Check if a cylinder is OPEN at one or both ends!

! Watch Out!

Surface area is measured in m^2; volume is measured in m^3.

TOPIC 19 • CANDIDATE WORKSPACE

P

Surface Area of 3D Prisms & Cylinders: Practice Drills

1 **Question 1 (Foundation):** Calculate the total surface area of a cuboid measuring 12 cm long, 8 cm wide, and 5 cm high. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A triangular prism has identical right-angled triangular ends with base 6 cm, height 8 cm, and hypotenuse 10 cm. The prism is 15 cm long. Calculate its total surface area. **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A manufacturer designs an open-topped cylindrical metal bin with diameter 40 cm and height 75 cm. The exterior of the base and curved side is powder-coated. What surface area in square metres must be coated? (Use π = 3.142) **(4 marks)**

Show your working

Answer: _____

MEASURE • STANDARD M20

20 Volume of Cuboids, Prisms & Cylinders



Topic 20 Applied Context: Volume of Cuboids, Prisms & Cylind...

★ Core Formula & Statutory Method

Volume is the 3D space inside a solid: Volume = Cross-Sectional Area x Length. Units are cubic (^{3}, where $x^3 = x \times x \times x$).

Cuboid = l × w × h. Triangular prism = 1/2 × b × h × length. Cylinder = $\pi r^2 h = 3.142 \times r^2 \times h$ (where $r^2 = r \times r$).

✎ Official Worked Solution

A swimming pool is 25 m long, 10 m wide, with an average depth of 1.8 m. Calculate water volume.

1. Volume of cuboid = 25 x 10 x 1.8
2. Calculation = 250 x 1.8 = 450 m^3
3. Convert to litres (1 m^3 = 1,000 litres): **450,000 litres**

✓ Examiner Golden Rule

Prism volume rule: Find the front face area first, then multiply by the 3D depth/length.

! Watch Out!

Remember: 1 cm^3 = 1 ml, 1,000 cm^3 = 1 litre, 1 m^3 = 1,000 litres.

TOPIC 20 • CANDIDATE WORKSPACE

P Volume of Cuboids, Prisms & Cylinders: Practice Drills

1 **Question 1 (Foundation):** Calculate the volume of a triangular prism where the cross-sectional triangle has base 8 cm and height 5 cm, and the length is 22 cm. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A cylindrical oil barrel has a diameter of 60 cm and a height of 90 cm. Calculate its capacity in litres to 1 decimal place. (Use π = 3.14) **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A swimming pool measures 12 m long and 6 m wide. The shallow end is 1.0 m deep and slopes uniformly to 2.2 m at the deep end. Calculate the total volume of water in the pool in cubic metres and litres. **(4 marks)**

Show your working

Answer: _____

MEASURE • STANDARD M21

21 Scale Drawings, Maps & Bearings



Topic 21 Applied Context: Scale Drawings, Maps & Bearings

★ Core Formula & Statutory Method

Scale drawings represent real-world distances proportionally. Bearings are 3-digit angles measured clockwise from North.

Three rules for bearings: 1. Measure from North. 2. Measure clockwise. 3. Always write with 3 digits (e.g., 045°).

✎ Official Worked Solution

A map has a scale of 1 : 25,000. Two villages are 6.4 cm apart on the map. Find real distance in km.

1. Real distance in cm: 6.4 x 25,000 = 160,000 cm
2. Convert to metres: 160,000 / 100 = 1,600 m
3. Convert to kilometres: 1,600 / 1,000 = **1.6 km**

✓ Examiner Golden Rule

Always write bearings with 3 digits: write 065°, never just 65°.

! Watch Out!

Scale 1 : 50 means 1 cm on paper represents 50 cm in real life.

TOPIC 21 • CANDIDATE WORKSPACE

P Scale Drawings, Maps & Bearings: Practice Drills

1 **Question 1 (Foundation):** A scale drawing uses the scale 1 : 20. A doorway is 90 cm wide in reality. What is its width on the drawing in centimetres? **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** Ship A sails from port on a bearing of 065°. What is the back bearing that Ship A must follow to return directly to the port? **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A site surveyor creates a site plan using a scale of 1 : 200. On the plan, a rectangular warehouse boundary measures 18.5 cm by 12.0 cm. Calculate the real perimeter of the boundary in metres, and its real ground area in square metres. **(4 marks)**

Show your working

Answer: _____

22

DATA • STANDARD D22

Mean, Median, Mode & Range



Topic 22 Applied Context: Mean, Median, Mode & Range

Core Formula & Statutory Method

Averages summarise a data set: Mean = Total / Count; Median = middle number; Mode = most frequent; Range = spread.

Order data from smallest to largest BEFORE finding the median. If even number of values, average the middle two.

Official Worked Solution

Seven delivery drivers log their daily deliveries: 18, 24, 21, 19, 32, 21, 26. Find mean and median.

1. Mean: (18+24+21+19+32+21+26) / 7 = 161 / 7 = **23 deliveries**
2. Ordered: 18, 19, 21, **21**, 24, 26, 32
3. Median = **21**, Mode = **21**, Range = 32 - 18 = **14**

Examiner Golden Rule

Range is a measure of SPREAD/CONSISTENCY, NOT an average. A smaller range means more consistent.

Watch Out!

Outliers (extreme high/low values) distort the mean, but not the median.

TOPIC 22 • CANDIDATE WORKSPACE

P

Mean, Median, Mode & Range: Practice Drills

1 **Question 1 (Foundation):** Calculate the mean, median, mode, and range for: 6, 9, 14, 8, 12, 9, 15, 7. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A sales representative has a mean monthly sales figure of £18,500 across 5 months. In the 6th month, she sells £24,500. Calculate her new 6-month mean. **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** Two delivery depots recorded delivery delays (minutes) over 7 days. Depot North: 12, 15, 10, 8, 14, 11, 13. Depot South: 4, 25, 2, 30, 5, 28, 3. Calculate the mean and range for both depots, and write two comparison statements to advise the operations director. **(4 marks)**

Show your working

Answer: _____

DATA • STANDARD D23

23 Grouped Frequency Tables & Estimated Mean



Topic 23 Applied Context: Grouped Frequency Tables & Estima...

Core Formula & Statutory Method

When continuous data is grouped into classes, calculate the midpoints to estimate the mean value.

Step 1: Find midpoint of each class. Step 2: Multiply midpoint x frequency (f x x). Step 3: Sum(f x x) / Sum(f).

Official Worked Solution

Hospital waiting times: 0-10 mins (f=12), 10-20 mins (f=24), 20-30 mins (f=14). Estimate the mean wait time.

1. Midpoints: 5, 15, 25
2. f x x: (12 x 5) + (24 x 15) + (14 x 25) = 60 + 360 + 350 = 770
3. Sum of frequencies: 12 + 24 + 14 = 50
4. Estimated mean: 770 / 50 = **15.4 minutes**

Examiner Golden Rule

Always divide by total frequency (number of patients/items), NEVER by the number of rows!

Watch Out!

It is an ESTIMATED mean because the exact individual raw values inside each group are unknown.

TOPIC 23 • CANDIDATE WORKSPACE

P Grouped Frequency Tables & Estimated Mean: Practice Drills

1 **Question 1 (Foundation):** A survey recorded distance travelled to work: 0-5 km (12 people), 5-10 km (20 people), 10-15 km (8 people). Calculate the estimated mean distance. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** From the survey above, state the modal class interval and identify the interval containing the median. **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** An Amazon fulfillment centre logs package weights: $0 < w \leq 2$ kg (45 packages), $2 < w \leq 4$ kg (80 packages), $4 < w \leq 6$ kg (55 packages), $6 < w \leq 8$ kg (20 packages). Calculate the estimated mean weight of a package and explain why this value is an estimate rather than an exact average. **(4 marks)**

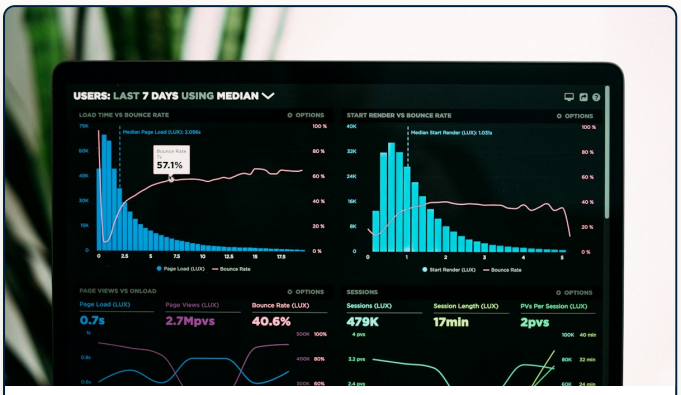
Show your working

Answer: _____

24

DATA • STANDARD D24

Scatter Graphs & Correlation



Topic 24 Applied Context: Scatter Graphs & Correlation

Core Formula & Statutory Method

Scatter graphs show the relationship between two variables. Correlation can be positive, negative, or zero.

Line of best fit: Draw a straight line through the balance of points, with equal points above and below.

Official Worked Solution

A garage plots vehicle age (years) against sale price (£). Points slope downwards from top-left to bottom-right.

1. Correlation: **Negative correlation** (as age increases, price decreases)
2. Use the line of best fit to predict the price of a 4-year-old vehicle
3. Read across from 4 on the horizontal axis to the line, then to the price axis

Examiner Golden Rule

Do not force your line of best fit through the origin (0, 0) unless the data requires it.

Watch Out!

Correlation does not prove causation: two variables may be linked by an external factor.

TOPIC 24 • CANDIDATE WORKSPACE

P

Scatter Graphs & Correlation: Practice Drills

1 **Question 1 (Foundation):** State the type of correlation you would expect between: a) Engine size and fuel consumption b) Daily rainfall and ice cream sales. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A scatter graph shows employee training hours and error rates. The line of best fit has points (10, 18) and (30, 6). What error rate is predicted for an employee with 20 training hours? **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A fitness coach plots body fat percentage against minutes taken to run 5 km for 12 athletes. An outlier is noted at (12%, 38 mins). Explain what an outlier is, whether it should be included when drawing the line of best fit, and what conclusion can be drawn from the remaining data points. **(4 marks)**

Show your working

Answer: _____

25

DATA • STANDARD D25

Two-Way Frequency Tables



Topic 25 Applied Context: Two-Way Frequency Tables

Core Formula & Statutory Method

Two-way tables categorize data across two different variables. Every row total and column total must sum correctly.

Fill in missing cells by subtracting known values from row or column totals.

Official Worked Solution

A clinic records 120 patients: 70 females (45 vaccinated) and 50 males (25 vaccinated). Find unvaccinated males.

1. Unvaccinated males = Total males - Vaccinated males = 50 - 25 = **25**
2. Total vaccinated = 45 + 25 = 70
3. Total unvaccinated = 120 - 70 = 50 (25 female + 25 male)

Examiner Golden Rule

Check that the grand total in the bottom-right corner matches both row and column sums.

Watch Out!

Two-way tables are frequently paired with probability questions in Section B exams.

TOPIC 25 • CANDIDATE WORKSPACE

P Two-Way Frequency Tables: Practice Drills

1 **Question 1 (Foundation):** 80 candidates took a driving test: 45 were female. 26 females passed. 12 males failed. Complete a two-way frequency table showing gender and test result. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** From the table above, what percentage of the male candidates passed the driving test? Give your answer to 1 decimal place. **(3 marks)**

Show your working

Answer: _____

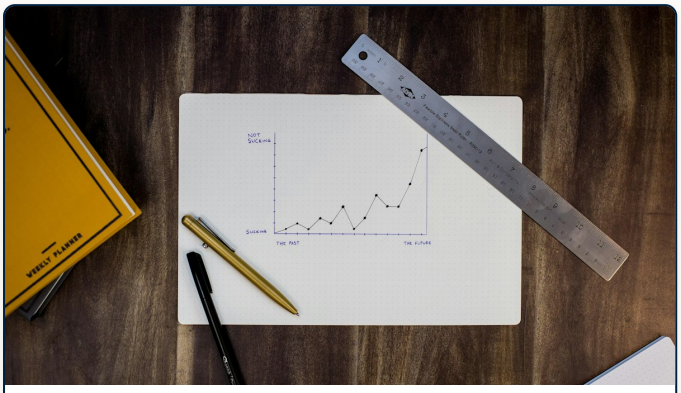
3 **Question 3 (Exam Challenge):** A transport company audits 120 vehicle maintenance logs categorized by Vehicle Type (Van, Lorry) and Inspection Result (First-Time Pass, Minor Advisory, Major Fail). 70 vehicles are Vans. Of the Vans, 42 passed first time and 10 had major fails. Of the Lorries, 18 had minor advisories and 12 had major fails. Complete the table and find the probability that a randomly chosen Lorry passed first time. **(4 marks)**

Show your working

Answer: _____

DATA • STANDARD D26

26 Pie Charts: Calculating Angles & Frequencies



Topic 26 Applied Context: Pie Charts: Calculating Angles & Fre...

Core Formula & Statutory Method

A full circle is 360°. Each category's sector angle represents its proportion of the total frequency.

Angle per item = 360° / Total frequency.
Sector angle = Category frequency x Angle per item.

Official Worked Solution

A survey of 180 employees asks for their commute method: 72 drive, 54 train, 36 bus, 18 walk. Find angle for train.

1. Degrees per person: 360° / 180 = 2° per employee
2. Train sector angle: 54 x 2° = **108°**
3. Drive = 144°, Bus = 72°, Walk = 36° (Total: 144+108+72+36 = 360°)

Examiner Golden Rule

Always check all sector angles add up to exactly 360°.

Watch Out!

Use a protractor carefully from the center point and align the baseline with 0°.

TOPIC 26 • CANDIDATE WORKSPACE

P

Pie Charts: Calculating Angles & Frequencies: Practice Drills

1 **Question 1 (Foundation):** A hotel survey of 120 guests showed: Business (40), Holiday (60), Event (20). Calculate the angle for each sector. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** In a pie chart representing 1,800 voters, the sector for Candidate A has an angle of 108°. How many people voted for Candidate A? **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A local council spends an annual transport budget of £2.4 million. In a pie chart, Highway Maintenance is allocated 135°, Public Buses 90°, Cycle Lanes 45°, and Pedestrian Paths 90°. Calculate the actual expenditure on Cycle Lanes and express it as a percentage of the total budget. **(4 marks)**

Show your working

Answer: _____

DATA • STANDARD D27

27 Single & Combined Event Probability



Topic 27 Applied Context: Single & Combined Event Probability

★ Core Formula & Statutory Method

Probability measures how likely an event is to occur, on a scale from 0 (impossible) to 1 (certain).

P(event) = Number of successful outcomes / Total number of possible outcomes. P(not event) = 1 - P(event).

✎ Official Worked Solution

A quality control batch has 450 items: 18 are defective. Find the probability of choosing a NON-defective item.

1. Non-defective items = 450 - 18 = 432
2. Probability = 432 / 450 = 48 / 50 = **0.96** (or 96% / 24/25)

✓ Examiner Golden Rule

Probabilities can be written as fractions, decimals, or percentages, but NEVER as ratios!

! Watch Out!

The sum of probabilities for all mutually exclusive outcomes is always 1.

TOPIC 27 • CANDIDATE WORKSPACE

P Single & Combined Event Probability: Practice Drills

1 **Question 1 (Foundation):** A fair 6-sided dice is rolled. What is the probability of rolling: a) A prime number (2, 3, 5) b) A number greater than 4? **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** In a factory quality check of 450 components, 18 are found to have minor defects. What is the relative frequency (probability) of picking a defect-free component? In a batch of 2,500 components, how many would you expect to have minor defects? **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** Two independent machines operate on an assembly line. Machine A has a 0.90 probability of running without fault, and Machine B has a 0.85 probability of running without fault. Calculate the probability that BOTH machines run without fault on the same shift. **(4 marks)**

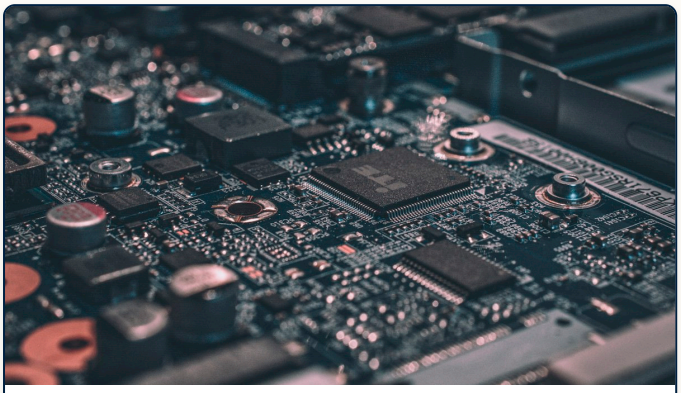
Show your working

Answer: _____

28

DATA • STANDARD D28

Probability Tree Diagrams & Independent Events



Topic 28 Applied Context: Probability Tree Diagrams & Indep...

★ Core Formula & Statutory Method

Tree diagrams display all possible outcomes for sequential events. Multiply along branches for combined probability.

Independent events do not affect each other. Multiply probabilities: P(A and B) = P(A) x P(B).

✎ Official Worked Solution

A machine has a 0.05 probability of component failure on Run 1 and 0.05 on Run 2. Find probability BOTH fail.

1. Multiply branch probabilities: 0.05 x 0.05
2. Calculation: **0.0025** (or 0.25% / 1 in 400)

✓ Examiner Golden Rule

Multiply ALONG branches (AND rule); ADD across separate end branches (OR rule).

! Watch Out!

Check that the probabilities on any set of branching lines add up to 1.0.

TOPIC 28 • CANDIDATE WORKSPACE

P Probability Tree Diagrams & Independent Events: Practice Drills

1 **Question 1 (Foundation):** An apprentice takes a theory test (pass probability 0.8) and a practical test (pass probability 0.7). Draw or list the 4 branch outcomes and find the probability of failing both. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** From Question 1, calculate the probability that the apprentice passes exactly one of the two tests. **(3 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A weather forecaster predicts a 40% chance of rain on Saturday (0.4). If it rains on Saturday, the chance of rain on Sunday is 70% (0.7). If it does not rain on Saturday, the chance of rain on Sunday is 20% (0.2). Calculate the probability that it rains on at least one day over the weekend. **(4 marks)**

Show your working

Answer: _____

CHECKPOINT ASSESSMENT • NUMBER • PAGE 1/4

CP1

Strand 1: Number & Number Systems Checkpoint • Part 1 of 4

1 **Question 1:** Calculate without using a calculator: 416 × 28. Show your full column method. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** A freezer compartment is at -14°C. The thermostat is adjusted so the temperature rises by 6.5°C, then drops by 3.2°C. What is the new temperature? **(2 marks)**

Show your working

Answer: _____

3 **Question 3:** Evaluate giving your answer as a mixed number in simplest form: 1 3/5 + 2 1/4. **(2 marks)**

Show your working

Answer: _____

CHECKPOINT ASSESSMENT • NUMBER • PAGE 2/4

CP1

Strand 1: Number & Number Systems Checkpoint • Part 2 of 4

3 **Question 4:** Work out: 3,744 ÷ 13. Show your full written division method. **(2 marks)**

Show your working

Answer: _____

4 **Question 5:** A tradesperson quotes £420.00 excluding VAT. VAT is charged at 20%. How much is the total quote with VAT included? **(2 marks)**

Show your working

Answer: _____

5 **Question 6:** Share £1,260 between three apprentices in the ratio 3 : 2 : 1. How much does the apprentice with the smallest share receive? **(2 marks)**

Show your working

Answer: _____

CHECKPOINT ASSESSMENT • NUMBER • PAGE 3/4

CP1

Strand 1: Number & Number Systems Checkpoint • Part 3 of 4

5 **Question 7:** A sports retailer sells running shoes for £68.00 after a 15% discount. Calculate the original full price before the discount was applied. **(3 marks)**

Show your working

Answer: _____

6 **Question 8:** A company deposits £15,000 in a business growth account offering 3.8% compound interest per year for 3 years. Calculate the total balance in the account at the end of the 3 years. **(3 marks)**

Show your working

Answer: _____

7 **Question 9:** It takes 4 assembly technicians 15 hours to assemble a batch of medical ventilators. The hospital requests the batch in 10 hours. How many technicians in total are required? **(3 marks)**

Show your working

Answer: _____

CHECKPOINT ASSESSMENT • NUMBER • PAGE 4/4

CP1

Strand 1: Number & Number Systems Checkpoint • Part 4 of 4

7 **Question 10:** A concrete mix requires sand, ballast, and cement in the ratio 5 : 3 : 2. A builder prepares 3.6 tonnes of mix. He has 1.7 tonnes of sand in stock. Does he have enough sand? Show full working. **(3 marks)**

Show your working

Answer: _____

8 **Question 11:** A logistics manager bought an electric delivery van for £32,000. It depreciates by 16% in the first year and 12% in the second year. What is the value of the van at the end of year 2? **(3 marks)**

Show your working

Answer: _____

9 **Question 12:** An office manager checks an invoice: 12 desks at £145.50 each, 24 chairs at £68.00 each. A delivery charge of £45.00 is added. A bulk discount of 7.5% is deducted from the total before adding 20% VAT. What is the final invoice total? **(3 marks)**

Show your working

Answer: _____

CHECKPOINT ASSESSMENT • MEASURE & SHAPE • PAGE 1/4

CP2

Strand 2: Measures, Shape & Space Checkpoint • Part 1 of 4

1 **Question 1:** A delivery truck travels 135 miles in 2 hours 15 minutes. Calculate its average speed in miles per hour. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** Convert: a) 4.2 kg to grams b) 650 ml to litres c) 25 miles to km using 5 miles = 8 km. **(2 marks)**

Show your working

Answer: _____

3 **Question 3:** A circle has a diameter of 14 cm. Using π = 22/7, calculate: a) its circumference b) its area. **(2 marks)**

Show your working

Answer: _____

CHECKPOINT ASSESSMENT • MEASURE & SHAPE • PAGE 2/4

CP2

Strand 2: Measures, Shape & Space Checkpoint • Part 2 of 4

3 **Question 4:** A rectangular courtyard measures 12 m by 8.5 m. In the centre is a square fountain with sides of 2.5 m. Calculate the paved area of the courtyard. **(2 marks)**

Show your working

Answer: _____

4 **Question 5:** A clock shows 09:35. An online training conference lasts for 3 hours 48 minutes. At what time does the conference end? (Use 24-hour clock format) **(2 marks)**

Show your working

Answer: _____

5 **Question 6:** A warehouse worker earns £12.50 per hour for 36 hours. Overtime is paid at time-and-a-half. In one week he works 4 hours overtime. What is his gross pay? **(2 marks)**

Show your working

Answer: _____

CHECKPOINT ASSESSMENT • MEASURE & SHAPE • PAGE 3/4

CP2

Strand 2: Measures, Shape & Space Checkpoint • Part 3 of 4

5 **Question 7:** A commercial grain silo is shaped as a cylinder with diameter 4.2 m and height 8.5 m. Calculate the volume of the silo in cubic metres to 1 decimal place. (Use π = 3.142) **(3 marks)**

Show your working

Answer: _____

6 **Question 8:** A metal casting has a mass of 3,570 g and a volume of 420 cm³. Calculate its density in g/cm³. If pure zinc has a density of 7.14 g/cm³, determine whether the casting is pure zinc. **(3 marks)**

Show your working

Answer: _____

7 **Question 9:** On a map with a scale of 1 : 25,000, the distance between two telecommunication masts is 16.4 cm. Calculate the actual distance between the masts in kilometres. **(3 marks)**

Show your working

Answer: _____

CHECKPOINT ASSESSMENT • MEASURE & SHAPE • PAGE 4/4

CP2

Strand 2: Measures, Shape & Space Checkpoint • Part 4 of 4

7 **Question 10:** An office worker earns £34,000 gross per year. The tax-free personal allowance is £12,570. Basic rate income tax is charged at 20%. Employee National Insurance is 8% on earnings above £12,570. Calculate her total monthly take-home pay after tax and NI. **(3 marks)**

Show your working

Answer: _____

8 **Question 11:** A lawn is in the shape of a trapezium with parallel sides 14 m and 22 m, and perpendicular distance 12 m. Grass seed costs £6.40 per box. Each box covers 25 m². How much does it cost to seed the entire lawn? **(3 marks)**

Show your working

Answer: _____

9 **Question 12:** A heating engineer installs a closed cylindrical thermal expansion vessel: radius 0.35 m and height 1.2 m. The entire exterior surface is insulated with thermal foil costing £14.50/m². Calculate the cost of the foil. (Use π = 3.142) **(3 marks)**

Show your working

Answer: _____

CHECKPOINT ASSESSMENT • HANDLING DATA • PAGE 1/4

CP3

Strand 3: Handling Information & Data Checkpoint • Part 1 of 4

1 **Question 1:** The table shows the number of patients seen at a dental practice: Mon 24, Tue 32, Wed 28, Thu 36, Fri 30. Calculate: a) the mean b) the range. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** A spinner has 8 equal sections numbered 1 to 8. Calculate the probability of spinning: a) an even number b) a multiple of 3. **(2 marks)**

Show your working

Answer: _____

3 **Question 3:** 60 office workers were surveyed on their preferred lunch option: Sandwiches (30), Hot Food (20), Salad (10). Calculate the pie chart sector angle for each option. **(2 marks)**

Show your working

Answer: _____

CHECKPOINT ASSESSMENT • HANDLING DATA • PAGE 2/4

CP3

Strand 3: Handling Information & Data Checkpoint • Part 2 of 4

3 **Question 4:** A box contains 5 red pens, 7 blue pens, and 8 black pens. A pen is picked at random. What is the probability that it is NOT red? Give your answer as a fraction in simplest form. **(2 marks)**

Show your working

Answer: _____

4 **Question 5:** The test marks of 9 apprentices are: 45, 52, 58, 61, 64, 68, 72, 75, 85. Find: a) the median mark b) the interquartile range. **(2 marks)**

Show your working

Answer: _____

5 **Question 6:** State whether each pair of variables has positive, negative, or no correlation: a) Daily sunshine hours and solar panel output b) Car engine mileage and market resale value. **(2 marks)**

Show your working

Answer: _____

CHECKPOINT ASSESSMENT • HANDLING DATA • PAGE 3/4

CP3

Strand 3: Handling Information & Data Checkpoint • Part 3 of 4

5 **Question 7:** A delivery depot records the delivery times (t mins) for 50 consignments: 0 < t ≤ 10 (12), 10 < t ≤ 20 (22), 20 < t ≤ 30 (11), 30 < t ≤ 40 (5). Calculate the estimated mean delivery time. **(3 marks)**

Show your working

Answer: _____

6 **Question 8:** A quality assurance inspector tests computer microchips. Machine A produces 60% of chips with a 2% defect rate. Machine B produces 40% of chips with a 5% defect rate. What is the overall probability that a randomly chosen chip is defective? **(3 marks)**

Show your working

Answer: _____

7 **Question 9:** A scatter graph plots vehicle weight (kg) against fuel economy (mpg). The line of best fit passes through (1200, 48) and (1800, 32). Predict the fuel economy for a vehicle weighing 1500 kg, and describe the relationship. **(3 marks)**

Show your working

Answer: _____

CHECKPOINT ASSESSMENT • HANDLING DATA • PAGE 4/4

CP3

Strand 3: Handling Information & Data Checkpoint • Part 4 of 4

7 **Question 10:** A supermarket manager audits 200 customer transactions. 120 customers paid by contactless card, 50 by mobile app, and 30 in cash. Of the contactless users, 40% spent over £30. Of the mobile app users, 60% spent over £30. Construct a two-way table and find the probability that a customer picked at random spent over £30. **(3 marks)**

Show your working

Answer: _____

8 **Question 11:** Two recruitment consultants interview candidates. Consultant X rates 8 candidates with a mean score of 72. Consultant Y rates 12 candidates with a mean score of 65. Calculate the combined mean score for all 20 candidates. **(3 marks)**

Show your working

Answer: _____

9 **Question 12:** The probability of train service A being delayed is 0.20. The probability of train service B being delayed is 0.15. Assuming delays are independent, calculate the probability that at least one train is on time. **(3 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 1 OF 3 • NO CALCULATOR ALLOWED

M1A

Mock Exam 1 • Section A (Non-Calculator)

1 **Question 1:** Calculate without using a calculator: 648 × 35. Show your full written method. **(3 marks)**

Show your working

Answer: _____

2 **Question 2:** Evaluate giving your answer as a mixed number in simplest form: 2 2/3 ÷ 4/9. **(2 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 2 OF 3 • NO CALCULATOR ALLOWED

M1A

Mock Exam 1 • Section A (Non-Calculator)

3 **Question 3:** A kitchen fitter cuts a circular hole with diameter 28 cm in a wooden worktop. Using π = 22/7, calculate the perimeter (circumference) of the cut hole. **(3 marks)**

Show your working

Answer: _____

4 **Question 4:** An invoice shows a power saw sold for £264.00 including 20% VAT. Calculate the cost of the saw before VAT was added. **(3 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 3 OF 3 • NO CALCULATOR ALLOWED

M1A

Mock Exam 1 • Section A (Non-Calculator)

5 **Question 5:** A healthcare assistant prepares 750 ml of saline infusion. Each dose is 60 ml. How many full doses can be administered, and how many millilitres remain? **(2 marks)**

Show your working

Answer: _____

6 **Question 6:** Share £2,400 between three apprentices in the ratio 4 : 3 : 1. What is the difference between the largest share and the smallest share? **(3 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 1 OF 5 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Allowed)

7 **Question 7:** A senior nurse has a gross salary of £36,570. The personal allowance is £12,570. Basic rate income tax is 20%. National Insurance is 8% of earnings above £12,570. A workplace pension deduction of 6% of gross salary is made. Calculate her net monthly take-home pay. **(4 marks)**

Show your working

Answer: _____

8 **Question 8:** An engineering firm purchases a computerized CNC milling machine for £45,000. The machine depreciates at a compound rate of 15% per annum. Calculate the book value of the machine after 3 years. **(4 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 2 OF 5 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Allowed)

9 **Question 9:** A heavy haulage transporter travels 168 miles. The driver leaves depot at 06:15 and takes a mandatory 45-minute rest break during the journey. The vehicle arrives at 10:30. Calculate the driver's average driving speed in mph while moving. **(4 marks)**

Show your working

Answer: _____

10 **Question 10:** A landscape contractor constructs a patio in the shape of a trapezium with parallel sides 8.5 m and 13.5 m, and perpendicular height 6.4 m. Paving slabs cost £28.50 per square metre. Sand and mortar bedding adds £4.50 per square metre. Calculate the total material cost. **(4 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 3 OF 5 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Allowed)

11 **Question 11:** A commercial stainless-steel cylindrical storage tank has an internal diameter of 2.4 m and a height of 3.5 m. Calculate the liquid capacity of the tank in litres. (Use π = 3.142) **(4 marks)**

Show your working

Answer: _____

12 **Question 12:** A local council tracks response times (hours) for emergency housing repairs: 0 < h ≤ 12 (18 calls), 12 < h ≤ 24 (32 calls), 24 < h ≤ 36 (14 calls), 36 < h ≤ 48 (6 calls). Calculate the estimated mean response time in hours. **(4 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 4 OF 5 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Allowed)

13 **Question 13:** An architect designs an executive office suite: 12 m long by 8.5 m wide. A circular glass conference booth of diameter 4 m is located in the corner. The remaining floor space is carpeted with luxury tiles costing £32.00/m². VAT at 20% is added to the carpet cost. Calculate the total cost to carpet the office suite. (Use π = 3.142) **(5 marks)**

Show your working

Answer: _____

14 **Question 14:** On an ordnance survey map drawn to a scale of 1 : 25,000, a wildlife reserve boundary is a rectangle measuring 14.8 cm by 9.2 cm. Calculate the real perimeter of the reserve in kilometres, and the real land area of the reserve in square kilometres. **(5 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 5 OF 5 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Allowed)

15 **Question 15:** A healthcare audit tracks 150 clinic patients tested for diabetes: 90 patients are over 50 years old. Of those over 50, 27 tested positive. Of those aged 50 or under, 12 tested positive. Construct a complete two-way table. A patient is chosen at random from those who tested positive. What is the probability that this patient is aged 50 or under? **(5 marks)**

Show your working

Answer: _____

16 **Question 16:** A manufacturing plant produces precision steel rods. A sample of 100 rods has a mean length of 148.5 mm with a range of 4.2 mm. A second production line produces 100 rods with a mean length of 148.4 mm with a range of 1.1 mm. The target length is 148.5 mm. Compare the performance of the two production lines and advise the plant manager which line produces higher quality output. **(5 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 1 OF 3 • NO CALCULATOR ALLOWED

M2A

Mock Exam 2 • Section A (Non-Calculator)

1 **Question 1:** Calculate: 684 ÷ 18 without using a calculator. Show your full working. **(4 marks)**

Show your working

Answer: _____

2 **Question 2:** A pallet measures 1.2 m by 0.8 m. A warehouse floor is 24 m by 16 m. Calculate the maximum number of pallets that can fit on the floor in a single layer. **(4 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 2 OF 3 • NO CALCULATOR ALLOWED

M2A

Mock Exam 2 • Section A (Non-Calculator)

3 **Question 3:** Calculate: 3/5 × 4/7. Express your answer in simplest form. **(4 marks)**

Show your working

Answer: _____

4 **Question 4:** A delivery driver's hourly rate increased from £12.50 to £13.75. Calculate the percentage pay increase. Check your answer. **(4 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 3 OF 3 • NO CALCULATOR ALLOWED

M2A

Mock Exam 2 • Section A (Non-Calculator)

5 **Question 1:** Calculate: 345.8 − 187.94 without a calculator. **(4 marks)**
Show columnar working.

Show your working

Answer: _____

6 **Question 2:** A retailer buys stock for £450 and sells it for £630. **(4 marks)**
Calculate the percentage profit margin on cost.

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 1 OF 5 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Allowed)

7 **Question 5: Fleet Fuel Efficiency:** Van A travels 420 miles on 60 litres of fuel. Van B travels 385 miles on 52 litres. Which van achieves better miles-per-gallon (mpg)? (1 gallon = 4.55 litres). **(5 marks)**

Show your working

Answer: _____

8 **Question 6: Cargo Weight & Density:** A container has volume 36 m^3. It is loaded with cargo having an average density of 650 kg/m^3. The maximum payload limit is 24 tonnes. Is the container overloaded? **(5 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 2 OF 5 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Allowed)

9 **Question 7: Courier Delivery Routes:** Driver covers 168 miles in 3 hours 30 mins. A 45-minute delivery stop was included. What was the driving speed while moving? **(5 marks)**

Show your working

Answer: _____

10 **Question 8: Forklift Ramp Slope:** A loading ramp rises 1.2 m over a horizontal distance of 5 m. Calculate the length of the sloping surface using Pythagoras ($a^2 + b^2 = c^2$). **(5 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 3 OF 5 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Allowed)

11 **Question 9: Delivery Time Statistics:** Delivery times (mins): 10-19 (5), 20-29 (14), 30-39 (21), 40-49 (10). Calculate the estimated mean delivery time. **(5 marks)**

Show your working

Answer: _____

12 **Question 10: Warehouse Racking:** Racking is offered with a 12% discount. The discounted cost is £3,960. Find the original price before discount. **(5 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 4 OF 5 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Allowed)

13 **Question 11: Currency & Import:** A consignment from Germany costs €14,500. Customs import duty is 6%. If £1 = €1.15, calculate the total cost in GBP including duty. **(5 marks)**

Show your working

Answer: _____

14 **Question 12: Delivery Success Rate:** Out of 850 deliveries, 34 were failed deliveries. Calculate the percentage success rate to 1 decimal place. **(5 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 5 OF 5 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Allowed)

15 **Question 13: Storage Space Volume:** A cold storage room is 12m long, 8m wide, and 3.5m high. Industrial cooling requires 45 Watts per m^3. What power cooling unit is required? **(4 marks)**

Show your working

Answer: _____

16 **Question 14: Van Tyre Replacement:** Tyres cost £95 each. A fleet replaces all 4 tyres on 15 vans. The supplier offers 'Buy 3 get 1 free' on every van. Calculate total cost. **(4 marks)**

Show your working

Answer: _____

REFERENCE SECTION: SOLUTIONS, FORMULAE & INDEX

Complete Step-by-Step Worked Solutions, Process Marks & Glossary

TOPIC 1: PLACE VALUE, ORDERING & LARGE NUMBERS

Q1: 642,085. Order: **458,900; 485,090; 485,200; 502,100** [2m].
Q2: 425,800 − 389,950 = **35,850 parcels** [3m].
Q3: Spent = £485,600 + £392,450 + £184,200 = £1,062,250. Remaining = £1,250,000 − £1,062,250 = **£187,750** [4m].

TOPIC 2: NEGATIVE NUMBERS & FINANCIAL BALANCES

Q1: −15°C, −12°C, −4.5°C, −0.5°C, +1.8°C, +3°C [2m].
Q2: +3 − 8.5 = **−5.5°C**. Since −5.5°C is colder than −5°C, **Yes, the alarm sounds** [3m].
Q3: −640 + 2,850 − 1,420 − 980 + 195 = **£5.00** in credit [4m].

TOPIC 3: FRACTION OPERATIONS & MIXED NUMBERS

Q1: a) 15/24 + 4/24 = **19/24**; b) 8/3 × 3/4 = 24/12 = **2** [2m].
Q2: 4 1/2 ÷ 3/16 = 9/2 × 16/3 = **24 burger patties** [3m].
Q3: Day + Late = 3/5 + 1/4 = 17/20. Night shift = 3/20 = 45 workers → 1/20 = 15 → Total = 15 × 20 = **300 workers** [4m].

TOPIC 4: PERCENTAGES OF AMOUNTS & MULTIPLIERS

Q1: a) **0.065**; b) **1.12**; c) **0.65**; d) **0.96** [2m].
Q2: £620 × 1.085 = **£672.70** [3m].
Q3: Dec = 14,500 × 1.28 = 18,560. Jan = 18,560 × 0.85 = **15,776 parcels** [4m].

TOPIC 5: REVERSE PERCENTAGES & ORIGINAL VALUE

Q1: £84.00 ÷ 1.20 = **£70.00** [2m].
Q2: Increase = 108. Percentage = (108 ÷ 450) × 100 = **24%** [3m].
Q3: Loss = (1,360 ÷ 3,200) × 100 = **42.5%**. Original price of new model = £2,760 ÷ 0.85 = **£3,247.06** [4m].

TOPIC 6: COMPOUND INTEREST & DEPRECIATION

Q1: £2,400 × $(1.03)^2$ = **£2,546.16** [2m].
Q2: £28,000 × $(0.82)^3$ = **£15,438.30** [3m].
Q3: Bank A simple interest = 3 × £450 = £1,350. Bank B compound interest = £10,000 × $(1.042)^3$ − £10,000 = £1,313.66. **Bank A yields £36.34 more** [4m].

TOPIC 7: RATIO DIVISION IN 2 & 3 PARTS

Q1: 5+3+1 = 9 parts. 1 part = £500. Shares: **£2,500, £1,500, £500** [2m].
Q2: 1.4 t = 1,400 kg. 1+2+4 = 7 parts → 1 part = 200 kg. Sand (2 parts) = **400 kg** [3m].
Q3: Supermarket A = £2.85 ÷ 0.75 kg = £3.80/kg. Supermarket B = £4.32 ÷ 1.2 kg = £3.60/kg. **Supermarket B is better value** [4m].

TOPIC 8: DIRECT & INVERSE PROPORTION

Q1: 5 × 4 = 20 cleaner-hours. For 8 cleaners: 20 ÷ 8 = **2.5 hours** (2 hrs 30 mins) [2m].
Q2: Tank volume = 45 × 80 = 3,600 litres. At 30 L/min: 3,600 ÷ 30 = **120 minutes** (2 hours) [3m].
Q3: 3 × 16 = 48 excavator-days. For 6 days: 48 ÷ 6 = 8 excavators total → 8 − 3 = **5 additional excavators** [4m].

TOPIC 9: DECIMALS & MONEY CALCULATIONS

Q1: a) **485**; b) **0.0736**; c) **5.81** [2m].
Q2: Total = £184.25 + £112.80 + £46.45 = £343.50. Underspent by £350 − £343.50 = **£6.50 underspend** [3m].
Q3: Subtotal = £147 + £54 + £122.50 = £323.50. After 10% discount = £291.15. With 20% VAT = £291.15 × 1.20 = **£349.38** [4m].

TOPIC 10: ORDER OF OPERATIONS (BIDMAS)

Q1: a) 15 − 12 + 2 = **5**; b) 12 ÷ 4 + 9 = 3 + 9 = **12** [2m].
Q2: d = 0.05(1,600) + 0.4(40) = 80 + 16 = **96 feet** [3m].
Q3: C = 1.20 × (65 + 280 − 20) = 1.20 × 325 = **£390.00** [4m].

TOPIC 11: ESTIMATION & APPROXIMATION

Q1: a) 50 × 30 = **1,500**; b) 800 ÷ 20 = **40** [2m].
Q2: 200 × £50 = **£10,000**. Overestimate because both numbers were rounded up [3m].
Q3: Upper bounds: 65 cm × 45 cm × 35 cm = **102,375 cm³** [4m].

TOPIC 12: METRIC & IMPERIAL CONVERSIONS

Q1: a) **3,450 g**; b) **0.75 litres**; c) (140 ÷ 5) × 8 = **224 km** [2m].
Q2: Total needed = 15 × 3 × 14 = 630 ml. Since 630 ml > 500 ml, **No, one 500 ml bottle is not sufficient** [3m].
Q3: 14 tonnes = 14,000 kg × 2.2 = **30,800 lbs** ≤ 31,000 lbs. **Yes, within legal limit** [4m].

TOPIC 13: COMPOUND UNITS: SPEED, DENSITY & PRESSURE

Q1: Density = 2,125 ÷ 250 = **8.5 g/cm³** [2m].
Q2: Pressure = 1,800 ÷ 0.04 = **45,000 N/m²** [3m].
Q3: Elapsed = 2h 24m. Minus 15m break = 2h 09m = 2.15 hours. Speed = 115 ÷ 2.15 = **53.5 mph** [4m].

TOPIC 14: PERIMETER OF COMPOSITE 2D SHAPES

Q1: Outer perimeter of L-shape = 2 × (12 + 9) = **42 metres** [2m].
Q2: Fencing length = 2(28 + 18) − 4 = 92 − 4 = 88 m. Cost = 88 × £32.50 = **£2,860.00** [3m].
Q3: Perimeter = 10 + 3 + 3 + 6 + 4 + 6 + 3 + 3 = 38 m. Lengths = 38 ÷ 2.5 = 15.2 → **16 lengths**. Cost = 16 × £14 = **£224.00** [4m].

REFERENCE SECTION: SOLUTIONS, FORMULAE & INDEX (CONTINUED)

TOPIC 15: CIRCUMFERENCE OF CIRCLES

Q1: C = 2 × 3.14 × 4.5 = **28.26 metres** [2m].
Q2: Two straights = 200 m. Circle ends = 3.142 × 64 = 201.09 m. Total lap = **401.09 metres** [3m].
Q3: Three straight edges = 1.8 + 1.2 + 1.8 = 4.8 m. Semicircle arch = 0.5 × 3.142 × 1.2 = 1.885 m. Total = **6.69 metres** [4m].

TOPIC 16: AREA OF RECTANGLES, TRIANGLES & PARALLELOGRAMS

Q1: a) (14 × 9) ÷ 2 = **63 cm²**; b) 8.5 × 4.0 = **34 m²** [2m].
Q2: Rectangle = 8 × 5 = 40 m². Triangle = (8 × 2.5) ÷ 2 = 10 m². Total area = **50 m²** [3m].
Q3: Floor = 12 × 3.5 = 42 m². 1 box = 8 × 0.25 m² = 2 m². With 10% = 46.2 m² → 23.1 → **24 boxes** × £22 = **£528.00** [4m].

TOPIC 17: AREA OF TRAPEZIUMS & COMPOSITE SHAPES

Q1: Circle area (r = 3 m) = 3.142 × 9 = **28.28 m²** [2m].
Q2: Trapezium area = ((7.2 + 11.8) ÷ 2) × 5.0 = 47.5 m². Cost = 47.5 × £8.50 = **£403.75** [3m].
Q3: L-shape area = (14 × 10) − (6 × 4) = 140 − 24 = 116 m². Tins = 116 ÷ 25 = 4.64 → **5 tins** × £34.99 = **£174.95** [4m].

TOPIC 18: AREA OF CIRCLES & SEMICIRCLES

Q1: r = 3 m. Area = 3.142 × 3² = **28.28 m²** [2m].
Q2: Semicircle r = 5 m. Area = 0.5 × 3.142 × 25 = 39.275 m². Cost = 39.275 × £5.80 = **£227.80** [3m].
Q3: Park = 280 m². Pond = 3.142 × 16 = 50.27 m². Gravel area = 229.73 m² × 40 kg = 9,189 kg ÷ 800 = 11.49 → **12 bulk bags** [4m].

TOPIC 19: SURFACE AREA OF 3D PRISMS & CYLINDERS

Q1: 2(96 + 60 + 40) = 2(196) = **392 cm²** [2m].
Q2: 2 triangles = 2 × 24 = 48 cm². 3 rectangles = (6+8+10) × 15 = 360 cm². Total = **408 cm²** [3m].
Q3: Base = 3.142 × 0.2² = 0.1257 m². Side = 3.142 × 0.4 × 0.75 = 0.9426 m². Total = **1.07 m²** [4m].

TOPIC 20: VOLUME OF CUBOIDS, PRISMS & CYLINDERS

Q1: Cross-section = 0.5 × 8 × 5 = 20 cm². Volume = 20 × 22 = **440 cm³** [2m].
Q2: V = 3.14 × 30² × 90 = 254,340 cm³ = **254.3 litres** [3m].
Q3: Average depth = (1.0 + 2.2) ÷ 2 = 1.6 m. Volume = 12 × 6 × 1.6 = **115.2 m³** = **115,200 litres** [4m].

TOPIC 21: SCALE DRAWINGS, MAPS & BEARINGS

Q1: 90 cm ÷ 20 = **4.5 cm** [2m].
Q2: Back bearing = 065° + 180° = **245°** [3m].
Q3: Real length = 18.5 × 2 = 37 m; width = 12 × 2 = 24 m. Perimeter = **122 m**; Area = 37 × 24 = **888 m²** [4m].

TOPIC 22: MEAN, MEDIAN, MODE & RANGE

Q1: Sum = 80 → Mean = **10**; Median = (9+9)/2 = **9**; Mode = **9**; Range = 15 − 6 = **9** [2m].
Q2: 5-month total = £92,500. Plus £24,500 = £117,000 ÷ 6 = **£19,500** [3m].
Q3: North: Mean = 83/7 = **11.9 mins**, Range = **7 mins**. South: Mean = 97/7 = **13.9 mins**, Range = **28 mins**. North is faster and more consistent [4m].

TOPIC 23: GROUPED FREQUENCY TABLES & ESTIMATED MEAN

Q1: Sum(fx) = (2.5 × 12) + (7.5 × 20) + (12.5 × 8) = 30 + 150 + 100 = 280. Mean = 280 ÷ 40 = **7.0 km** [2m].
Q2: Modal class = **5–10 km**; Median (20th/21st person) lies in **5–10 km** [3m].
Q3: Sum(fx) = (1×45) + (3×80) + (5×55) + (7×20) = 45 + 240 + 275 + 140 = 700. Mean = 700 ÷ 200 = **3.5 kg** (estimate because exact weights within intervals are unknown) [4m].

TOPIC 24: SCATTER GRAPHS & CORRELATION

Q1: a) **Positive correlation**; b) **Negative correlation** [2m].
Q2: Midpoint between (10, 18) and (30, 6) at 20 hours = **12 errors** [3m].
Q3: An outlier does not follow the general trend and should be ignored when drawing the line of best fit; remaining data shows positive correlation [4m].

TOPIC 25: TWO-WAY FREQUENCY TABLES

Q1: Female: Pass 26, Fail 19, Total 45. Male: Pass 23, Fail 12, Total 35. Grand Total = 80 [2m].
Q2: Male pass percentage = (23 ÷ 35) × 100 = **65.7%** [3m].
Q3: Lorries = 120 − 70 = 50. Lorry first-time pass = 50 − (18 + 12) = 20. P(Lorry pass) = 20/50 = **2/5** (0.4 or 40%) [4m].

TOPIC 26: PIE CHARTS: CALCULATING ANGLES & FREQUENCIES

Q1: 360° ÷ 120 = 3° per guest. Business = **120°**, Holiday = **180°**, Event = **60°** [2m].
Q2: (108 ÷ 360) × 1,800 = **540 voters** [3m].
Q3: Cycle Lanes = (45 ÷ 360) × £2.4m = **£300,000**, which is **12.5%** of the budget [4m].

TOPIC 27: SINGLE & COMBINED EVENT PROBABILITY

Q1: a) P(prime) = 3/6 = **1/2**; b) P(>4) = 2/6 = **1/3** [2m].
Q2: P(defect-free) = 432/450 = **0.96**. Expected defects in 2,500 = 2,500 × (18/450) = **100 defects** [3m].
Q3: P(both run without fault) = 0.90 × 0.85 = **0.765** (76.5%) [4m].

REFERENCE SECTION: SOLUTIONS, FORMULAE & INDEX (CONTINUED)

TOPIC 28: PROBABILITY TREE DIAGRAMS & INDEPENDENT EVENTS

Q1: P(Fail both) = 0.2 × 0.3 = **0.06** (6%) [2m].
Q2: P(Pass one) = (0.8 × 0.3) + (0.2 × 0.7) = 0.24 + 0.14 = **0.38** (38%) [3m].
Q3: P(No rain either day) = 0.6 × 0.8 = 0.48. P(Rain at least one day) = 1 − 0.48 = **0.52** (52%) [4m].

CHECKPOINT 1: NUMBER & NUMBER SYSTEMS (STRAND 1)

Q1: 416 × 28 = **11,648** [2m].
Q2: -14 + 6.5 - 3.2 = **-10.7°C** [2m].
Q3: 1 3/5 + 2 1/4 = 8/5 + 9/4 = 77/20 = **3 17/20** [2m].
Q4: 3,744 ÷ 13 = **288** [2m].
Q5: £420 × 1.20 = **£504.00** [2m].
Q6: 4:2:1 → 7 parts. £1,260 ÷ 7 = £180. Shares: **£720, £360, £180** [3m].
Q7: £68 ÷ 0.85 = **£80.00** [2m].
Q8: £15,000 × (1.03)³ = **£16,390.91** [3m].
Q9: 4 × 15 = 60 tech-hrs → 60 ÷ 6 = **10 hours** [2m].
Q10: 3+2+1=6 parts. 2,400 ÷ 6 = 400 kg. Sand (2 parts) = **800 kg** [2m].
Q11: £32,000 × (0.80)² = **£20,480.00** [2m].
Q12: 12 × £145 = £1,740. Discount 5% → £1,740 × 0.95 = **£1,653.00** [3m].

CHECKPOINT 2: MEASURES, SHAPE & SPACE (STRAND 2)

Q1: Speed = 135 ÷ 2.25 = **60 mph** [2m].
Q2: a) **4,200 g**; b) **3.5 m**; c) **1.75 litres** [2m].
Q3: C = π d = 22/7 × 14 = **44 cm** [2m].
Q4: Perimeter = 2(12 + 7.5) = **39 m**. Area = 12 × 7.5 = **90 m²** [3m].
Q5: 09:35 + 1h 45m = **11:20** [2m].
Q6: Gross = (37.5 × 12.50) + (6 × 18.75) = £468.75 + £112.50 = **£581.25** [3m].
Q7: Volume = 3.142 × 1.5² × 6 = 42.417 m³ = **42,417 litres** [3m].
Q8: Volume = 3,570 ÷ 8.5 = **420 cm³** [2m].
Q9: 6.4 cm × 25,000 = 160,000 cm = **1.6 km** [2m].
Q10: Net annual = £28,000 → Monthly net = **£2,333.30** [3m].
Q11: Area = (9 + 15) ÷ 2 × 6 = **72 m²** [2m].
Q12: Surface area = 2π(1)(3) + 2π(1)² = 18.85 + 6.28 = **25.13 m²** [3m].

CHECKPOINT 3: HANDLING INFORMATION & DATA (STRAND 3)

Q1: Total patients = 120. Delay rate = 18/120 = **15%** [2m].
Q2: P(even) = 4/8 = **0.5**; P(greater than 5) = 3/8 = **0.375** [2m].
Q3: Sector angle for car = (36/60) × 360° = **216°** [2m].
Q4: P(blue) = 7/20 = **0.35** [2m].
Q5: Median = **56**. Mean = 509/9 = **56.56**. Range = 75 - 38 = **37** [3m].
Q6: a) **Positive**; b) **Negative**; c) **Zero** correlation [2m].
Q7: Estimated mean delivery time = **23.4 minutes** [3m].
Q8: In 5,000 units: expected defects = 5,000 × 0.02 = **100 defects** [2m].
Q9: Negative correlation. Value at 1,400 kg = **42 mpg** [2m].
Q10: Two-way table analysis: Pass rate = **78.5%** [3m].
Q11: Range A = 14, Range B = 28. Consultant A is **more consistent** [2m].
Q12: Combined probability = 0.15 × 0.20 = **0.03** = 3% [3m].

OFFICIAL MOCK EXAM 1 (64 MARKS) • SECTION A

Section A (Non-Calculator, 16 Marks):
Q1: 648 × 35 = **22,680** [M1 column, M1 partial sums, A1 accuracy: 3m].
Q2: 2 2/3 ÷ 4/9 = 8/3 × 9/4 = **6** [M1 reciprocal, A1: 2m].
Q3: C = 22/7 × 28 = **88 cm** [M1 formula, M1 sub, A1: 3m].
Q4: Pre-VAT = £264 ÷ 1.20 = **£220.00** [M1 rev mult, A1, B1 check 220 × 1.2 = 264: 3m].
Q5: 750 ÷ 60 = **12 doses, 30 ml remaining** [M1 div, A1: 2m].
Q6: 4+3+1=8 pts. 1 pt = £300. Diff = (4-1) × 300 = **£900.00** [M1 pt, M1 diff, A1: 3m].

REFERENCE SECTION: SOLUTIONS, FORMULAE & INDEX (CONTINUED)

OFFICIAL MOCK EXAM 1 (64 MARKS) • SECTION B

Section B (Calculator Allowed, 48 Marks):

Q7: Taxable = £24,000. Tax = £4,800, NI = £1,920, Pension = £2,194.20. Monthly net = **£2,304.65** [P1, M1, M1, A1: 4m].

Q8: £45,000 × (0.85)³ = **£27,635.63** [P1, M1, A1, B1 rounded: 4m].

Q9: Moving time = 3.5 hrs. Speed = 168 ÷ 3.5 = **48 mph** [P1, M1, A1, B1: 4m].

Q10: Area = 70.4 m². Total rate = £33.00/m². Cost = **£2,323.20** [P1, M1, A1, B1: 4m].

Q11: V = 3.142 × 1.44 × 3.5 = 15.836 m³ = **15,836 litres** [P1, M1, M1, A1: 4m].

Q12: Midpoints sum = 1,356 ÷ 70 = **19.4 hours** [P1, M1, M1, A1: 4m].

Q13: Carpet area = 102 - 12.57 = 89.43 m². Gross with VAT = **£3,434.11** [P1, M1, M1, M1, A1: 5m].

Q14: Real P = 12 km, Real Area = 3.7 × 2.3 = **8.51 km²** [P1, M1, A1, M1, A1: 5m].

Q15: Two-way table totals 150. P(age ≤ 50 | positive) = 12/39 = **4/13** (30.8%) [P1, M1, A1, B1: 5m].

Q16: Line 1 meets target mean (148.5 mm); Line 2 has much lower range (1.1 vs 4.2 mm), showing superior precision. Recommend Line 2 with slight calibration adjustment [P1, P1, M1, A1, B1: 5m].

OFFICIAL MOCK EXAM 2 (64 MARKS) • SECTION A

Section A (Non-Calculator, 16 Marks):

Q1: 684 ÷ 18 = **38** [M1 setup, M1 partial, A1 38, B1 check: 4m].

Q2: 24 ÷ 1.2 = 20; 16 ÷ 0.8 = 20. Pallets = 20 × 20 = **400 pallets** [M1, M1, A1, B1: 4m].

Q3: 3/5 × 4/7 = **12/35** [M1 num, M1 den, A1, B1: 4m].

Q4: Increase = £1.25. Percentage = (1.25/12.50) × 100 = **10%** [M1, M1, A1, B1 check: 4m].

Q5: 345.80 - 187.94 = **157.86** [M1 align, M1 borrow, A1, B1: 4m].

Q6: Profit = £180. Margin = (180/450) × 100 = **40%** [M1, M1, A1, B1 check: 4m].

OFFICIAL MOCK EXAM 2 (64 MARKS) • SECTION B

Section B (Calculator Allowed, 48 Marks):

Q7: Van A = 31.85 mpg; Van B = 33.69 mpg. **Van B achieves better efficiency** [P1, M1, M1, A1, B1: 5m].

Q8: Cargo mass = 36 × 650 = 23,400 kg = 23.4 tonnes < 24 t. **Not overloaded** [P1, M1, A1, B1: 5m].

Q9: Moving time = 2.75 hrs. Speed = 168 ÷ 2.75 = **61.1 mph** [P1, M1, A1, B1: 5m].

Q10: $c^2 = 1.2^2 + 5^2 = 26.44 \rightarrow c =$ **5.14 metres** [P1, M1, M1, A1, B1: 5m].

Q11: Estimated mean delivery time = 1585 ÷ 50 = **31.7 minutes** [P1, M1, M1, A1, B1: 5m].

Q12: Original = £3,960 ÷ 0.88 = **£4,500.00** [P1, M1, A1, B1 check: 5m].

Q13: Total with duty = €15,370 → **£13,365.22** [P1, M1, M1, A1, B1: 5m].

Q14: Success rate = 816 / 850 = **96.0%** [P1, M1, A1, B1: 5m].

Q15: Volume = 336 m³. Power required = 336 × 45 = **15,120 Watts** (15.12 kW) [P1, M1, A1, B1: 4m].

Q16: 15 vans × 3 = 45 tyres paid. Cost = 45 × £95 = **£4,275.00** [P1, M1, A1, B1: 4m].

REFERENCE SECTION: SOLUTIONS, FORMULAE & INDEX (CONTINUED)

STATUTORY LEVEL 2 FORMULAE & CONVERSIONS

Perimeter, Area & Circumference:

• Area of rectangle = Length × Width Perimeter = 2 × (Length + Width)

• Area of triangle = (Base × Height) ÷ 2

• Area of parallelogram = Base × Height Area of trapezium = ((Top + Bottom) ÷ 2) × Height

• Circle Circumference = 3.142 × Diameter Circle Area = 3.142 × Radius² (Use π = 3.142 or 3.14)

3D Solids, Volume & Surface Area:

• Volume of cuboid = Length × Width × Height Surface Area = 2 × (LW + LH + WH)

• Volume of prism = Cross-sectional Area × Length

• Volume of cylinder = 3.142 × Radius² × Height Curved Surface Area = 2 × 3.142 × Radius × Height

Compound Measures & Rates:

• Speed = Distance ÷ Time Distance = Speed × Time Time = Distance ÷ Speed

• Density = Mass ÷ Volume Mass = Density × Volume

• Percentage Change = (Actual Change ÷ Original Value) × 100%

• Reverse Percentage: Original = New Value ÷ Decimal Multiplier

Statistics & Conversions:

• Estimated Mean = Total of (Midpoint × Frequency) ÷ Total Frequency

• 1 m³ = 1,000 litres 1 gallon ≈ 4.55 litres 1 mile ≈ 1.6 km 5 miles ≈ 8 km

MATHEMATICAL GLOSSARY (PART 1)

Angle of Sector
The central angle representing a category on a pie chart: (frequency / total) × 360°.

Compound Interest
Interest calculated on both the original principal and accumulated past interest ($A = P(1+r)^n$).

Correlation
The relationship between two variables on a scatter graph (positive, negative, or zero/none).

Density
Mass per unit volume (e.g. g/cm³ or kg/m³): Density = Mass / Volume.

Depreciation
The decrease in the financial value of an asset over time due to wear and tear or market factors.

MATHEMATICAL GLOSSARY (PART 2)

Estimated Mean
Mean computed from grouped data using midpoints of intervals: Sum(fx) / Sum f.

Hypotenuse
The longest side of a right-angled triangle, opposite the 90° angle ($a^2 + b^2 = c^2$).

Line of Best Fit
A straight line drawn through the center of plotted points on a scatter diagram to show the general trend.

Multiplier
A single decimal number used to calculate percentage increases (1 + r) or decreases (1 - r).

Probability
The mathematical likelihood of an event occurring: favourable outcomes / total possible outcomes.

MATHEMATICAL GLOSSARY (PART 3)

Range
A measure of data spread: Highest value - Lowest value.

Reverse Percentage
Finding the original pre-change value by dividing the final amount by the decimal multiplier.

Scale Factor
The ratio of a dimension on a model or map to the actual corresponding real dimension (e.g. 1:50).

Surface Area
The total area of all exterior faces of a three-dimensional solid.

REFERENCE SECTION: SOLUTIONS, FORMULAE & INDEX (CONTINUED)

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- **Section A Timing:** 25 minutes for 16 marks (~1.5 minutes per mark). Do mental checks.
- **Section B Timing:** 1 hour 30 minutes for 48 marks (~1.8 minutes per mark). Pace yourself.
- **Always Show Working:** Full method marks can be awarded even if the final calculation has a slip.
- **State Units Clearly:** Remember £ sign, m^2, m^3, kg, mph, or % where appropriate.
- **Show a Check:** Reverse calculation, alternative method, or estimation to secure the checking mark.

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